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Pragmatic Processes in Survey Interviewing

Frederick G. Conrad, Michael F. Schober, and Norbert Schwarz

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Abstract and Keywords

In interpreting survey questions, respondents rely as much on the pragmatics of everyday communication as they do on interpreting syntax or literal meaning. Respondents make inferences about the intended meaning of questions based on a large array of contextual cues, from judgments about the researcher to their sense of how common ground builds up in ordinary conversation and their perception of the actions of interviewers (or automated interviewing systems). How respondents answer questions—the speed and fluency of their answers or requests for clarification—is informative about the accuracy and reliability of their answers. Although survey respondents try to make sense of questions using whatever conversational resources they can bring to bear, researchers are often insensitive to these processes and surprised by their impact.

Keywords: pragmatics, survey, questions, dialogue, conversation, inferences, accuracy, clarification, interviewer, automated system

When people fill out online customer satisfaction questionnaires, respond to political polls on the telephone, talk to census enumerators, or participate in laboratory research that involves questionnaires, they are engaged in a complex communicative act that involves more processes than are immediately apparent. Of course, they must comprehend the literal meaning of the words in the questions, retrieve relevant facts and opinions, and formulate their answers. But they also make inferences about the intentions behind the questions based on a large array of contextual cues—what is ordinarily considered to belong to the domain of linguistic *pragmatics*, as opposed to semantics or syntax. Some of these inferences are made on the basis of prior knowledge that respondents apply to the survey situation, and may make in real time while hearing or reading questions. For example, respondents may use their knowledge about the questioner's affiliation, and therefore likely intentions, or knowledge about what speakers usually mean when they ask more than one question on the same topic. Other inferences are made on the basis of subsequent turns of dialogue, whether this “dialogue” is with a human interviewer or an automated interviewing system. For example, a respondent may alter his or her interpretation of what a word or phrase in a question means after receiving clarifying information or even a verbatim repetition of the question from an interviewer.

In this essay, we review evidence about these kinds of pragmatic processes in survey interviews. We argue that they occur because survey respondents bring their ordinary conversational practices to survey interviews, even though surveys are a specialized form of interaction (see, e.g., Houtkoop-Steenstra, 2000; Schaeffer, 1991; Schober, 1999; Schober & Conrad, 2002; Schwarz, 1994). In everyday conversations, people have an array of tools available to bridge the gaps between what they mean and what listeners take them to mean. In general, they assume (p. 421) that they are engaged in a cooperative endeavor with their interlocutor—that they are following what Grice (1975) called the *cooperative principle*. So listeners try to make sense of what has been said, even if the literal content doesn't make sense; because they assume that a speaker is trying to communicate something, they infer an intended meaning, even if that meaning is indirect, ironic, or hostile. As we will see, in surveys, this

can lead respondents to make unwarranted inferences in assuming that the formal features of questionnaires have been designed following that cooperative principle, even though they often have not (Schwarz, 1994, 1996). In typical conversations, people also assume that they have the ability to *ground* their understanding with their partner (Clark, 1996; Clark & Schaefer, 1989; Clark & Wilkes-Gibbs, 1986)—to engage in additional turns of clarification dialogue to make sure they have understood each other sufficiently for their current purposes. When survey respondents are confronted with interviewing situations that explicitly prevent grounding dialogue, they can end up interpreting survey questions differently than researchers intend and ultimately provide inaccurate answers without knowing it.

Surveys as an Unusual Conversational Situation

Survey interactions are unusual on several fronts. First, whether the survey is administered by an interviewer or self-administered (for example a paper-and-pencil questionnaire), the primary interlocutors are the respondent and an absent researcher or researching agency who have designed the questions (or commissioned someone else to design them). In an interviewer-administered survey, there is an intermediary, the interviewer, who presents the questions on behalf of the researchers and facilitates the data collection by keeping the interview on track, motivating the respondent, and recording the answers. In self-administered surveys, there may be no extra party involved in the “dialogue” between the researchers and the respondent, although modern technologies are blurring the distinction between interviewer- and self-administered surveys as they embed features of human interviewing, such as interactive dialogue, into their systems.¹

Second, even when an interviewer is present in a survey interview, the goal of standardized survey interviewing is to minimize any potentially biasing influence of the interviewer on the responses. A set of standardized practices has evolved to reduce the likelihood that an individual interviewer will bias answers (see Fowler & Mangione, 1990, for a review of the state of the art and the underlying rationale; see also Beatty, 1995). The key features are that interviewers read questions exactly as worded, probe neutrally, and never allow their own ideas to influence the respondents’ answers. Observance of these methodological rules is assumed to reduce or eliminate what survey researchers refer to as *interviewer-related error*—any systematic effect of a particular interviewer on survey responses.

Third, the requirement of standardized interviewers to administer only “neutral” or nondirective probes creates an unusual dynamic in the interaction (see Schaeffer, 1991; Schober & Conrad, 2002; Suchman & Jordan, 1990, 1991). Consider an interview in which the interviewer presents the question “How many hours per week do you usually work?” If a respondent asks the interviewer for clarification—“Should I consider business lunches to be work?”—she is unlikely to get a satisfying answer under standardized interviewing conditions. To prevent the interviewer’s providing additional substantive information to this respondent that would not be available to a respondent who didn’t inquire, an interviewer following strictly standardized procedures is required to reply with a neutral probe such as rereading the question (“Let me repeat the question: How many hours per week do you usually work?”) or indicating that the interpretation of the question is the sole obligation of the respondent (e.g., “Whatever it means to you”). This kind of move would be extremely strange in an everyday conversation, and, even if respondents tend to be aware that they are in an unusual interactive situation, there is evidence that this kind of interaction can be frustrating (Suchman & Jordan, 1990, 1991). Aside from frustration, this leads to the unusual feature that, in survey interviews, interviewers relinquish control over what the question means. In most conversations, speakers are responsible for the meaning of what they say (Clark, 1996; Clark & Schober, 1992), and they usually have the ability to engage in additional dialogue to clarify and verify—to ground—what was intended.

Nonetheless, survey interactions do share many other important features with everyday conversations, most notably the back-and-forth structure of question-asking and answering and the reliance on additional contextual information to make sense of what is being said. We argue that it is exactly because people do not (perhaps cannot) turn off their ordinary reasoning and meaning-making when participating in a survey interview that the pragmatics of (p. 422) survey interviewing are particularly important to understand.

Inferring Meaning from Features of the Questionnaire

The constraints that standardized communication impose on the mutual negotiation of meaning force participants to rely on other information to make sense of the researcher's questions. To do so, they make extensive use of contextual information, going beyond the literal meaning of the question asked to infer what the question designers intended, as licensed by Grice's (1975, 1978) cooperative principle. Listeners make sense of utterances on the assumption that speakers are following (at least) four of what Grice called *maxims*. Listeners assume that speakers make their contribution such that it can be understood by their audience, not only avoiding ambiguity and wordiness, but also taking the characteristics of their audience into account, designing their utterance in a way that the audience can figure out what they mean (*Grice's maxim of manner*). Central to this process are speakers' assumptions about the information that they share with recipients—that is, the common ground (Schiffer, 1972; Stalnaker, 1978); listeners interpret the speaker's utterance against what they assume to constitute the common ground (e. g., Clark, Schreuder, & Buttrick, 1983). Moreover, listeners assume that each successful contribution to the conversation extends the common ground of the participants, in line with their assumption that speakers are trying to make all contributions relevant to the aims of the ongoing conversation (*Grice's maxim of relation*). Listeners thus use the context of an utterance to disambiguate its meaning by making bridging inferences (Clark, 1977); they assume speakers will not make a contribution to a conversation that is irrelevant to its goal, unless it is marked as such. Listeners assume that speakers make their contribution as informative as is required, but not more informative than is required (*Grice's maxim of quantity*), without reiterating information that the recipient already has or may take for granted anyway (Clark & Haviland, 1977). Finally, listeners assume that speakers do not say things they believe to be false or lack adequate evidence for (*Grice's maxim of quality*).

As Grice (1975) noted, these maxims apply most directly to situations in which people attempt to exchange information to get things done. Of course, many conversations are characterized by other goals, such as entertaining one another (see Higgins, Fondacaro, & McCann, 1982), and such additional goals can be at play in survey interviews, as interviewers work to build rapport and motivate respondents. Nonetheless, the assumptions underlying the cooperative principle remain at play even when additional goals are present, in a number of ways.

Researcher's Affiliation

In natural conversations, listeners take their knowledge about the speaker into account when they interpret his or her utterances. So do research participants. For example, Norenzayan and Schwarz (1999) asked respondents to explain a case of workplace homicide, described in a newspaper clipping. The otherwise identical questionnaire was either printed on the letterhead of an "Institute for Personality Research" or on the letterhead of an "Institute for Social Research." As expected, respondents' open-ended explanations entailed a greater emphasis on personality variables or on social-contextual variables, depending on whether they thought the researcher was a personality psychologist or a social scientist. Similarly, Galesic and Tourangeau (2007) asked respondents otherwise identical questions about workplace behavior as part of an alleged "Sexual Harassment Survey" conducted for Women Against Sexual Harassment or as part of an alleged "Work Atmosphere Survey" conducted for a Work Environment Institute. Respondents perceived the same behaviors as more likely to represent sexual harassment when they were presented as part of a sexual harassment survey—after all, why else would they have been included?

Such findings highlight that participants take the researchers' likely epistemic interest into account to provide an answer that is relevant to the goals of the ongoing conversation, much as they would be expected to in daily life. Beyond simply providing the kind of answer that they believe the researcher wants to hear, respondents can perceive the domain of inquiry differently depending on the questioner's identity and thus his or her presumed research goals. Unfortunately, researchers themselves are often insensitive to the information that their institutional affiliation or the title of their study may convey.

Preceding Questions

In natural conversations, listeners draw on the already established common ground when they interpret a speaker's utterances. In stark contrast, researchers would often prefer respondents to (p. 423) answer each question in isolation, thus avoiding the emergence of any order effects. Respondents, however, may not always be aware that the usual rules of conversational conduct do not apply. Hence, they do their best to make sense of the questions asked by drawing on the context in which they are presented, including the content of preceding

questions.

Suppose, for example, that respondents are asked questions about highly obscure or even completely fictitious issues, such as the “Agricultural Trade Act of 1978” (e.g., Bishop, Oldendick, & Tuchfarber, 1986; Schuman & Presser, 1981). In a typical study, some 30 percent of the respondents of a representative sample are likely to report an opinion on these issues, despite the fact that they can hardly know anything about it. Some researchers attributed this to respondents’ “fear of appearing uninformed,” which may induce them “to conjure up opinions even when they had not given the particular issue any thought prior to the interview” (Erikson, Luttborg, & Tedin, 1988, p. 44), casting serious doubt on the meaningfulness of survey responses. From a conversational perspective, however, respondents merely do what they would be expected to do in any other conversation: they make sense of an ambiguous utterance by drawing on its context. Testing the possibility that context created by prior questions can affect interpretation of subsequent questions, Strack, Schwarz, and Wänke (1991, experiment 1) asked German college students about their attitude toward the introduction of an “educational contribution” allegedly discussed in state parliament. For some participants, this question was preceded by a question about the average tuition fees that students have to pay at US universities (in contrast to Germany, where university education is free), whereas others had to estimate how much money Swedish students receive from their government as financial support. At the end of the study, participants were asked what the “educational contribution” implied. As expected, they inferred that students would receive money when the fictitious issue was preceded by the Swedish support question, but that students would have to pay money when it was preceded by the American tuition question. Not surprisingly, they favored the introduction of an “educational contribution” in the former case, but opposed it in the latter case.

Open Versus Closed Question Formats

Respondents also rely on contextual information when the question is less obviously ambiguous than in the just presented case. Suppose, for example, that you are asked to report what you have done today. Most likely, you would not include in your report that you took a shower, that you dressed, and so on. However, if these activities were included in a list of response alternatives, you would probably endorse them. This thought experiment reflects a robust set of findings (see Schwarz & Hippler, 1991, for a review): any given opinion or behavior is less likely to be volunteered in an open response format (the respondent answers in his or her own words) than to be endorsed in a closed response format (multiple choice options), if presented. On the other hand, opinions or behaviors that are omitted from the set of response alternatives in a closed format are unlikely to be reported at all, even if an “other” category is explicitly offered, which respondents in general rarely use. Several processes contribute to this reliable pattern.

Most importantly, respondents are unlikely to spontaneously report, in an open answer format, information that seems self evident or irrelevant, as in the shower example just given. In refraining from these responses, they follow the conversational maxim that an utterance should be informative and should provide the information that the recipient is interested in. This results in an underreporting of presumably self evident information that is eliminated by closed response formats, in which the explicit presentation of the proper response alternative indicates the investigator’s interest in this information. Similarly, a generic “other” response provides little information and would be considered inadequate as an answer in most conversations; hence, it is rarely checked. In addition, the response alternatives may remind respondents of options that they may otherwise not have considered.

The Numeric Values of Rating Scales

When constructing a rating scale, researchers typically pay attention to the verbal end anchors used and to the number of scale points presented (see Dawes & Smith, 1985, for a review). Having settled for a 7-point rating scale, for example, they are less likely to worry whether those seven points should be represented by unnumbered boxes, by numbers ranging from 1 to 7, or by numbers ranging from –3 to +3. However, the specific numerical values used may again change the inferred meaning of the question, strongly affecting the obtained responses.

For example, Schwarz, Knäuper, Hippler, Noelle-Neumann, and Clark (1991a, experiment 1) asked a representative sample of German (p. 424) adults, “How successful would you say you have been in life?” This question was accompanied by an 11-point rating scale, with the endpoints labeled “not at all successful” and “extremely

successful.” In one condition, the numeric values of the rating scale ranged from 0 (“not at all successful”) to 10 (“extremely successful”), whereas in the other condition they ranged from –5 (“not at all successful”) to +5 (“extremely successful”). The results showed a pronounced impact of the numeric values used. Whereas 34 percent of the respondents endorsed a value between 0 and 5 on the 0–10 scale, only 13 percent endorsed one of the formally equivalent values between –5 and 0 on the –5 to +5 scale.

As subsequent experiments showed (Schwarz et al., 1991a), respondents draw on the numeric values of the rating scale to interpret the verbal anchor “not at all successful”—does this term refer to the absence of noteworthy success or to the presence of explicit failure? When the numeric values of the scale run from 0 to 10, respondents assume that the researcher has a unipolar dimension in mind, in which different values reflect different degrees of the same attribute. Conversely, when the numeric values run from –5 to +5, they infer that the researcher has a bipolar dimension in mind, in which one endpoint (–5) refers to the opposite of the other (+5). Accordingly, “not at all successful” refers to the mere absence of success when combined with the numeric value “0” but to the presence of failure when combined with the numeric value “–5.”

Respondents’ use of numeric values in making sense of verbal labels is not restricted to variations that include or omit negative numbers. For example, Schwarz, Grayson, and Knäuper (1998) asked undergraduates how often they engaged in a variety of low-frequency activities. In all conditions, the 11-point rating scale ranged from “rarely” to “often.” However, “rarely” was combined with the numeric value 0 or 1. As expected, respondents interpreted “rarely” to mean “never” when combined with 0, but to mean “a low frequency” when combined with 1. As a result, they provided higher mean frequency ratings along the 0–10 ($M = 2.8$) than the 1–11 scale ($M = 1.9$; scale recoded to 0–10).

Frequency Scales

Research participants are often asked to report the frequency with which they engage in a behavior by checking the appropriate value from a set of frequency response alternatives provided to them (see Table 26.1 for an example). Again, the range of response alternatives may serve as a source of information for respondents (see Schwarz, 1999b, for a review). In general, respondents assume that researchers construct a meaningful scale that reflects appropriate knowledge about the distribution of the behavior. Accordingly, values in the middle range of the scale are assumed to reflect the “average” or “typical” behavior, whereas the extremes of the scale are assumed to correspond to the extremes of the distribution. These assumptions influence respondents’ interpretation of the question, their behavioral reports, and related judgments.

Table 26.1 Reported daily TV consumption as a function of response alternatives

Low Frequency		High Frequency	
Alternatives		Alternatives	
Up to 1/2 h	7.4%	Up to 2 1/2h	62.5%
1/2 h to 1h	17.7%	2 1/2h to 3h	23.4%
1h to 1 1/2h	26.5%	3h to 3 1/2h	7.8%
1 1/2h to 2h	14.7%	3 1/2h to 4h	4.7%
2h to 2 1/2h	17.7%	4h to 4 1/2h	1.6%
More than 2 1/2h	16.2%	More than 4 1/2h	0.0%

Question Interpretation

Suppose, for example, that respondents are asked to indicate how frequently they were “really irritated” recently. To give an informative answer, they must decide what the researcher means by “really irritated”: does the term refer to major or minor irritations? When the scale presents high-frequency response alternatives, participants may infer that the researcher is interested in relatively frequent events, whereas low-frequency response alternatives presumably indicate an interest in rare events. Given that major irritations are less frequent than minor ones, the range of frequency values may therefore indirectly specify the kind of irritations the researcher has in mind.

Empirically, this is the case. For example, Schwarz, Strack, Müller, and Chassein (-1988) asked participants to report how frequently they were “really irritated” along a high-frequency scale ranging from “several times daily” to “less than once a week” or along a low-frequency scale ranging from “several times a year” to “less than once every three months.” Subsequently, participants reported a typical example of irritation. As expected, their examples were more extreme when the preceding question presented a low- rather than high-frequency scale, indicating that they drew on the frequency scale to determine the intended meaning of the question. Accordingly, identical question stems in combination with different frequency scales result in different interpretations and hence assess different experiences (see Schwarz, 1999, for a review).

Frequency Estimates

Even if the behavior under investigation is reasonably well defined, however, the range of response alternatives may strongly affect respondents’ frequency estimates. Mundane behaviors of a high frequency, such as watching TV, for example, are not represented in memory as distinct episodes; instead, the individual episodes blend into generic (p. 425) representations that lack time and space markers (see Belli, 1998, for a discussion). Accordingly, respondents cannot recall and count relevant episodes but have to resort to various estimation and inference strategies to arrive at a plausible answer (for more detailed treatments, see Brown, 2002; Schwarz & Oyserman, 2001). One of these strategies draws on the range of the frequency scale as a frame of reference, resulting in higher frequency estimates along scales that present high rather than low frequency response alternatives.

For example, only 16.2 percent of a sample of German consumers reported watching TV for more than 2.5 hours a day when presented with the low-frequency scale shown in Table 26.1, whereas 37.5 percent did so when presented with the high-frequency scale (Schwarz, Hippler, Deutsch, & Strack, 1985). Similar results have been obtained for a wide range of different behaviors, including health behaviors (e.g., Schwarz & Scheuring, 1992), sexual behaviors (e.g., Tourangeau & Smith, 1996), and consumer behaviors (e.g., Menon, Raghubir, & Schwarz, 1995). More demanding estimation tasks increase the degree to which scale values affect participants’ personal frequency estimates (e.g., Bless, Bohner, Hild, & Schwarz, 1992), and, the less concrete information people have in memory, the more they rely on the scale when making their judgments (e.g., Schwarz & Bienias, 1990). In contrast, the impact of response alternatives is weak or absent when the question pertains to highly regular behaviors (e.g., brushing teeth), for which respondents can draw on rate-of-occurrence information (e.g., “twice a day”; Menon, 1994; Menon et al., 1995).

Comparative Judgments

In addition, the frequency range of the response alternatives has been found to affect subsequent comparative judgments. Given the assumption that the scale reflects the distribution of the behavior, checking a response alternative is the same as locating one’s own position in the distribution. Accordingly, respondents extract comparison information from their own location on the response scale and use this information in making subsequent comparative judgments.

For example, checking 2h on the low-frequency scale shown in Table 26.1 implies that a respondent’s TV consumption is above average, whereas checking the same value on the high-frequency scale implies that his or her TV consumption is below average. As a result, respondents in the Schwarz et al. (1985) studies reported that TV plays a more important role in their leisure time (experiment 1) and described themselves as less satisfied with the variety of things they do in their leisure time (experiment 2) when they had to report their TV consumption on the low- rather than the high-frequency scale. Moreover, these frame of reference effects are not limited to respondents themselves, but influence the users of their reports as well. For example, in a study by Schwarz, Bless, Bohner, Harlacher, and Kellenbenz (1991, experiment 2), experienced medical doctors considered having

the same physical symptom twice a week to reflect a more severe medical condition when “twice a week” was a high rather than a low response alternative on the symptoms checklist presented to them.

Reference Periods

Similarly, Winkielman, Knäuper, and Schwarz (1998) observed that the length of the reference period can profoundly affect question interpretation. In their studies, respondents were either asked how frequently they had been angry “last week” or “last year.” Again, they inferred that the researcher is (p. 426) interested in more frequent and less severe episodes of anger when the question pertained to one week rather than one year, and their examples reflected this differential question interpretation.

On theoretical grounds, one may further expect that formal features of questionnaires, like the values of a frequency scale or the length of a reference period, seem more relevant when they are unique to the question asked rather than shared by many heterogeneous questions. In the latter case, respondents may conclude that this is the format used for all questions, rendering it less informative for the intended meaning of any given one. Empirically, this is the case. For example, using the same reference period for several substantively unrelated questions attenuates its influence on question interpretation relative to conditions in which each question is associated with a unique reference period (Igou, Bless, & Schwarz, 2002).

Summary

As the reviewed examples illustrate, minor variations in question wording or format can profoundly affect respondents’ answers (Schwarz, 1999a). They do so because respondents draw on the context in which a question is asked to infer its intended meaning. In most surveys, respondents have no other choice because there is either nobody to ask for clarification (as in the case of self-administered questionnaires, mail, and most web surveys) or a well-trained standardized interviewer will refuse to provide additional information. This reliance on contextual information is licensed by the cooperative principle (Grice, 1975); respondents merely do what they would be expected to do in everyday interactions. Unfortunately, their efforts are rarely appreciated by most researchers, who consider formal characteristics of questionnaires and the content of adjacent questions substantively irrelevant and treat their influence as an undesirable artifact. In most studies, researchers are unlikely to even note such influences, given that most surveys only ask questions in one way, without experimentally manipulating question format.

Giving an Informative Answer: Question Repetition and Meaning Change

The preceding examples illustrated how research participants draw on conversational maxims in interpreting the researcher’s contributions (i.e., the questions). Not surprisingly, participants also heed the maxims that underlie their conduct of conversation in daily life when they design their own answer to the researcher’s questions. Consistent with the maxim of quantity, they observe the common ground and try to provide information that is new to the researcher, rather than information they have already given earlier—or information the researcher may take for granted anyway.

As an example, consider a study in which respondents were asked to report their marital satisfaction, as well as their general life-satisfaction in different orders (Schwarz, Strack, & Mai, 1991b). As shown in the first column of Table 26.2, the answers to both questions correlated $r = .32$ when the general life-satisfaction question preceded the more specific marital satisfaction question. Reversing the question order increased this correlation to $r = .67$. This reflects that answering the marital satisfaction question brought marriage-related information to mind that respondents might otherwise not have considered. This interpretation is supported by a correlation of $r = .61$ when a reworded version of the general question explicitly asked respondents to take their marriage into account.

Table 26.2 Correlation of relationship satisfaction and life-satisfaction as a function of question order and conversational context

	Number of Specific Questions	
	One	Three
Condition		
General-specific	.32*	.32*
Specific-general	.67*	.46*
Specific-general,		
with joint lead-in	.18	.48*
Specific-general,		
explicit inclusion	.61*	.53*
Specific-general,		
explicit exclusion	.20	.11

N = 50 per cell, except for "Specific-general with joint lead-in," N = 56. Correlations marked by an asterisk differ from chance, $p < .05$. Adapted from Schwarz, N., Strack, F., & Mai, H. P. (1991). Assimilation and contrast effects in part-whole question sequences: A conversational logic analysis. *Public Opinion Quarterly*, 55, 3–23. Reprinted by permission.

In a third condition, Schwarz and colleagues deliberately evoked the conversational norm of nonredundancy (maxim of quantity). To do so, they introduced both questions with a joint lead-in that (p. 427) read, "We now have two questions about your life. The first pertains to your marital satisfaction and the second to your general life-satisfaction." Under this condition, the same question order that resulted in $r = .67$ without a joint lead-in now produced a low and nonsignificant correlation of $r = .18$. This suggests that respondents deliberately ignored information that they had already provided in response to a specific question when making a subsequent general judgment. Apparently, they interpreted the general question as if it were worded, "Aside from your marriage, which you already told us about, how satisfied are you with other aspects of your life?" Consistent with this interpretation, a correlation of $r = .20$ was obtained when the general question was reworded in this way.

Note, however, that the pragmatic implications of the norm of nonredundancy may change when several specific questions precede a more general one. Suppose, for example, that respondents are asked to report on their job satisfaction, their leisure time satisfaction, and their marital satisfaction before a general life-satisfaction question is presented. In this case, they may interpret the general question in two different ways. On the one hand, they may assume that it is a request to consider still other aspects of their life, much as if it were worded, "Aside from what you already told us,..." On the other hand, they may interpret the general question as a request to integrate the previously reported aspects into an overall judgment, much as if it were worded, "Taking these aspects together, ..." This interpretational ambiguity does not arise if only one specific question is asked, as in this example. In that case, an interpretation of the general question in the sense of "taking all aspects together" would make little sense because only one aspect was addressed, thus rendering this interpretation of the general question completely redundant with the specific one. If several specific questions are asked, however, both interpretations are viable. In this case, an integrative judgment is informative because it does provide "new" information about the relative importance of the respective domains. Moreover, "summing up" at the end of a series of related thoughts is

acceptable conversational practice, whereas there is little to sum up if only one thought was offered.

To explore these possibilities, other respondents were asked three specific questions pertaining to their leisure time satisfaction, their job satisfaction, and, finally, their marital satisfaction. As shown in the second column of Table 26.2, the correlation between marital satisfaction and life satisfaction increased from $r = .32$ to $r = .46$ when answering the specific questions first brought information about one's marriage to mind. However, this increase was less pronounced than when the marital satisfaction question was the only specific question that preceded the general one ($r = .67$), reflecting that the three specific questions brought a more varied set of information to mind. More importantly, introducing the three specific questions and the general one by a joint lead-in did *not* reduce the resulting correlation, $r = .48$. This indicates that respondents adopted a "Taking-all-aspects-together" interpretation of the general question if it was preceded by three, rather than one, specific question(s). This interpretation is further supported by a correlation of $r = .53$ when the general question was reworded to request an integrative judgment and a correlation of $r = .11$ when the reworded question required the consideration of other aspects of one's life.

In combination, these findings further illustrate that the interpretation of an identically worded question may change as a function of conversational variables, resulting in markedly different responses. Moreover, the resulting differences are not restricted to the means or margins of the response distribution, as social scientists have frequently assumed. Rather, context variables may result in different correlational patterns, thus violating the assumption that context effects would be restricted to differences in the means, whereas the relationship between variables would be "form resistant" (Schuman & Duncan, 1974). Obviously, we would draw very different substantive conclusions about the contribution of marital satisfaction to overall life satisfaction, depending on the order in which these questions are presented and the perceived conversational context evoked by their introduction.

Adjusting Interpretation in Survey Dialogue

Pragmatic processes in survey interviewing are at play not only in respondents' initial interpretation of questions, but also in the interaction—the conversational turns or subsequent respondent behaviors with an automated system—that follows the presentation of the questions. In those rare interviews that allow clarification dialogue, respondents can adjust their interpretation of questions from definitions that interviewers (or automated interviewing systems) give or instructions about the response task (e.g., the degree of precision required in an answer); clarification can be given either as a result of the respondent's direct request (p. 428) for help ("what do you mean by 'work for pay'?") or because the interviewer has diagnosed the respondent's need for additional help from the respondent's tentative-sounding answer ("uh...yes?"). In a strictly standardized interview, respondents cannot rely on explicit clarification or help, but they still can adjust their interpretations on the basis of interviewer actions and what those actions might mean in an everyday conversation.

It is worth highlighting that more goes on in survey dialogue than one would think from simply looking at an interviewer's script. Beyond interviewers asking question and respondents immediately providing answers followed immediately by the next question, interviewers often accept answers ("got it," "okay") in ways that can inform respondents about the suitability of their answers and potentially about the interviewer's evaluation of the content of the answers, although interviewers are trained to avoid giving evaluative feedback on answers. Interviewers can repeat a respondent's answer to verify that they heard it correctly ("three times a week?")—which has the potential to sound like a challenge or surprise at the answer. Just as almost any feature of a questionnaire can provide information about what a researcher might have intended, there is virtually nothing an interviewer can say or do that doesn't have the potential to influence a respondent's interpretation—nodding or *not* nodding, writing down an answer or *not* writing down an answer, smiling or *not* smiling, all have the potential to be treated as meaningful by respondents and to be used to infer researchers' intentions.

Inferring Interpretation in Strictly Standardized Interviews

As we noted earlier, strictly standardized interviewers are trained to behave in ways that would be quite unusual in an everyday conversation. This can limit the kinds of conversational moves they can make in ways that have the potential to seriously affect survey data quality.

In particular, respondents in strictly standardized interviews cannot engage in the kind of clarification dialogue that would help align their interpretation of individual words with the intentions of the question authors. And the likelihood of misalignment can be much greater than one might think; respondents' interpretations of ordinary terms in even well-prettested questions can be surprisingly variable. For example, Belson (1981) reported that respondents' interpretation of "reading a magazine" covered a wide range of different behaviors, from having seen the magazine at a newsstand to having read it cover-to-cover or having subscribed to it. Similarly, Suessbrick, Schober, and Conrad (2000) asked respondents, "Have you smoked at least 100 cigarettes in your entire life?" Whereas 54 percent of the respondents interpreted this as including "cigarettes that you took a puff or two from," 23 percent thought it referred to "cigarettes you partially smoked" (more than a puff or two but not finished), and 23 percent interpreted it as including "only cigarettes that you finished." There was a similar range in whether respondents counted only tobacco cigarettes, herbal and marijuana cigarettes, pipes, cigars, or chewing tobacco; whether they counted cigarettes that they smoked without inhaling; and whether they counted only cigarettes they themselves had bought.

Respondents' interpretations were variable enough that when they were reasked the same question, asking them to include any cigarettes that they took even one puff of but only tobacco cigarettes, 10 percent of the respondents changed their answers, indicating that their initial interpretation of the question was not shared with the researchers or with each other. Their initial interpretation also had the problematic effect of leading them down different paths of the survey, and thus answering quite different subsequent questions, than they would have with a different starting interpretation. This kind of conceptual variability in interpretation can thus harm survey estimates for more than just a single question.

As this example makes clear, the nature of ambiguity in surveys goes beyond simple lexical ambiguity, which usually refers to words having different senses (e.g., "bank" as a financial institution vs. edge of a river). The ambiguity in the smoking example has to do with the range of activities that a respondent includes in the term—which may not be stable across all different moments of use of "smoking" in a respondent's life or day because people may have different judgments about what is relevant to include in different conversations. In surveys that ask respondents about their own lives, there is an additional layer of ambiguity having to do with the mapping between the researcher's concept, the respondent's (chronic or momentary) concept, and the respondent's actual life circumstances. For example, a respondent who has never touched a cigarette or one who smokes tobacco cigarettes all the time may conceive of smoking rather differently than the researcher, but their answer to the smoking question will be unaffected by the difference. It is (p. 429) only for respondents who have smoked occasionally and whose interpretation of smoking differs from the researcher's that the misalignment can lead to answers that are inconsistent with researcher's concepts and thus problematic survey estimates.

As another example, consider these questions from major surveys of US government agencies. "During the past year, have you purchased or had expenses for household furniture?" If you bought an end table, you shouldn't have much trouble answering "Yes." But what if you bought a floor lamp—does that qualify as "furniture"? Or consider the question, "Last week, how many hours did you work?" If you have a 9-to-5 job, the meaning may not be ambiguous, but what if you do business over lunch or solve work-related problems while jogging? How about an academic who keeps up with developments in her field by reading at night? As the examples illustrate, many questions have a high degree of what we have called *mapping ambiguity*; that is, the mapping between the question term and the respondent's circumstances remains unclear.

As we have noted, the ideal of strictly standardized interviewing is to leave the interpretation of survey terms entirely up to respondents and to deal with any requests for clarification with neutral probes that return the responsibility for interpretation to the respondent. Nonetheless, interviewers' neutral probes can (intentionally or not) direct respondents to particular interpretations. Consider this example from a telephone interview (Schober & Conrad, 1997), in which the interviewer repeats the question with contrastive stress on the critical word ("COLLEGE") in a way that clearly informs the respondent about the researcher's interpretation of college tuition:

INTERVIEWER (I): Has Alexander purchased or had expenses for college tuition or fixed fees?

RESPONDENT (R): Um, he's got...um tuition for secretarial school.

I: Pardon me, I ...I didn't...

R: He has tuition for secretarial school.

I: Has Alexander purchased or had expenses for COLLEGE tuition?

R: No.

I: Or fixed fees?

R: No.

Here, the interviewer has followed the letter of the law for standardized interviewing but has nonetheless provided additional information to this respondent (and probably not to others), at odds with the goals of neutral probing. The respondent has used his knowledge of the pragmatics of dialogue to determine the researcher's interpretation in a way that would be invisible to the researcher without close examination of the interview dialogue.

Clarification Dialogue in Surveys

Although strictly standardized interviews try to prevent the use of everyday techniques for grounding understanding, one could imagine administering surveys in ways that encourage interviewers to use nonstandardized wording to help respondents interpret survey questions more uniformly, in keeping with respondents' (and interviewers') everyday conversational practices. This approach has been explored in a series of studies by Conrad, Schober, and colleagues that compare response accuracy and interaction patterns in strictly standardized and more collaborative "conversational" interviews, as well as for a range of interviewing techniques in between. These studies have produced consistent findings across a range of survey modes involving human interviewers (telephone and face-to-face) and automated interviewing systems (textual web surveys, virtual animated interviewers, speech interactive voice response [IVR]) that allow clarification in different ways.

In one set of studies, professional interviewers telephoned naïve respondents either in the laboratory (Schober & Conrad, 1997; Schober, Conrad, & Fricker, 2004) or at home (Conrad & Schober, 2000) and asked questions from large US government surveys. In the laboratory studies, respondents answered the questions on the basis of scenarios that described the work, housing, and purchases of fictional people. Because the researchers created the scenarios, they knew the correct answers—according to official definitions—for each question-scenario combination, allowing the determination of response accuracy. In the household study, respondents reported about their own lives, and response accuracy was evaluated by less direct reinterview methods.

In all interviews, questions were first posed exactly as worded, essentially standardizing the "starting point" under both standardized and conversational interviewing approaches; the difference between approaches concerns whether it is ultimately wording or underlying meaning that is standardized. In the more collaborative interviews, interviewers were (p. 430) then encouraged to ground understanding of question meaning. Across the experimental conditions in the different studies, they were trained to do this either by providing clarification only when respondents explicitly requested it, as in this example (from Schober & Conrad, 1997):

I: Has Kelly purchased or had expenses for household furniture?

R: Um...is a lamp furniture?

I: No sir, we do not include lamps and lighting fixtures.

R: Okay, no.

or by also providing clarification when, in their judgment, respondents seemed to need it (based on how they had answered or not answered the question), as in this example (Schober, Conrad, & Fricker, 2004):

I: How many hours per week does Mindy usually work at her job?

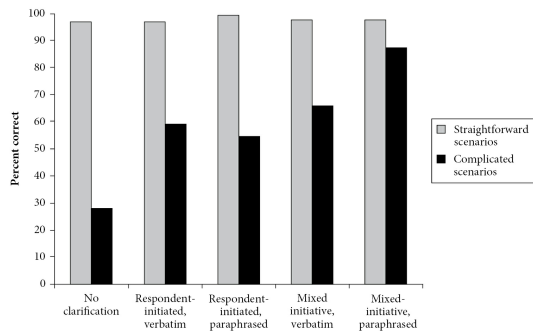
R: (long pause)

I: And by usually I mean fifty percent of the time or more, or the most frequent schedule during the past four or five months.

R: Uh, fifty hours.

I: Okay.

In some conditions they were trained to provide full verbatim definitions when they provided clarification, and, in others, they could paraphrase what they deemed to be the most relevant parts of definitions. In all studies, all interviewers learned the official definitions of key concepts in the questions, whether or not they were to provide definitions in the interview, so that differential interviewer knowledge could not account for any effects on response accuracy.



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Fig. 26.1 Response accuracy for straightforward and complicated scenarios across five interviewing conditions: (1) no clarification (standardized), (2) respondent-initiated clarification provided verbatim, (3) respondent-initiated clarification that interviewers could paraphrase, (4) mixed initiative (respondent- or interviewer-initiated) provided verbatim, and (5) mixed initiative (respondent- or interviewer-initiated) clarification that interviewers could paraphrase.

Across all studies with well-specified fictional scenarios (Schober & Conrad, 1997; Schober, Conrad, & Fricker, 2004), a reliable pattern emerged (see Figure 26.1). When the mappings between question concepts and people's circumstances (scenarios) were straightforward, all interviewing techniques led to nearly perfect accuracy; virtually all respondents interpreted question concepts in the ways that survey designers intended. But when mappings between question concepts and people's circumstances were complicated, response accuracy was lowest (28 percent) under strictly standardized interviewing conditions (Schober & Conrad, 1997). Response accuracy improved dramatically when interviewers provided clarification on request (54 and 59 percent in Schober, Conrad, & Fricker, 2004, experiment 1), but not as much as when they could also give clarification when they thought (p. 431) it might help (87 percent in Schober & Conrad, 1997). Whether interviewers' clarifications were scripted (i.e., they provided definitions verbatim) or unscripted (i.e., they could paraphrase definitions) (Schober, Conrad, & Fricker, 2004, experiment 1), did not really matter. Even when interviewers were merely instructed to do what they ordinarily do, they sometimes provided clarification (Schober, Conrad, & Fricker, 2004, experiment 2), and, in these cases, response accuracy was much higher (79 percent) than when they left the interpretation of questions entirely up to respondents (23 percent).

These results were mirrored in a household study (Conrad & Schober, 2000) in which participants reported on their own lives. In this study, respondents' actual circumstances were unknown, and response accuracy was assessed through a reinterview; the initial interview was standardized and the second interview was conversational for half the respondents and standardized for the other half. Respondents changed their answers more when their second interview was conversational (22 percent) than when it was strictly standardized (11 percent), a change that must be attributed to the fact that the clarification available in the conversational reinterview led to a reinterpretation of the questions consistent with the researchers' intended interpretations. Consistent with this finding, when asked about their purchases, respondents were more likely to answer on the basis of "legal" purchases (i.e., those that fit the definition of the purchase category) when the second interview was conversational (95 percent) than when it was standardized (57 percent). For example, in conversational interviews, respondents who answered "yes" to a question about moving expenses were more likely to do so because they were considering expenses included in the definition (e.g., they had hired commercial movers) than in standardized interviews, where they were more likely to respond "yes" because of excluded expenses (e.g., do-it-yourself moving). This could clearly be attributed to interviewers' explaining what to count and what not to count (i.e., grounding the meaning of the term). The improved response accuracy in conversational interviews suggests that respondents' actual circumstances

(as opposed to the fictional scenarios presented in the lab studies) are complicated sufficiently often—at least for these questions—to justify a collaborative approach to survey interviewing.

Despite the improved accuracy that comes with conversational interviewing and the feasibility of the technique for use by ordinary telephone interviewers, the benefits come at an operational cost. Conversational interviews took substantially longer than standardized interviews (from 80 to 300 percent longer) depending on the number of complicated mapping situations. It is difficult to determine how much of a cost issue this is for surveys because the frequency of complicated mappings across a survey population and for a particular topic domain is usually unknown, and conversational partners (here, survey interviewers and respondents) often don't recognize that they might have misaligned interpretations (Schober, 2005).

The findings from these studies replicate in studies of surveys using automated interviewing agents, as opposed to human interviewers. Using the same fictional scenarios from the Schober and Conrad (1997) study, Conrad, Schober, and Coiner (2007) demonstrated that laboratory respondents in a self-administered web (browser-based) survey that allowed respondents to click for clarification of survey terms provided more accurate responses when answering about complicated mapping scenarios than did respondents who were unable to click for clarification. Respondents answered even more accurately for complicated mapping scenarios when the automated system could also volunteer clarification even if the respondent hadn't explicitly requested it, based on long delays before responding. Similarly, Ehlen et al. (2007) showed that respondents in an automated speech dialogue survey system who could request clarification or who were provided with clarification when they took too long to answer provided more accurate responses for complicated mapping scenarios than did respondents who did not have access to clarification dialogue.

Conversational interviewing seems to improve comprehension in much the same way, not only in surveys with nonsensitive fact-based questions but also when respondents answer survey questions about their opinions, where it is less immediately evident what an accurate answer would be. As Hubbard, Antoun, and Conrad (2012) demonstrated, conversational interviewing improved response accuracy for attitude questions about drowsy driving (e.g., "In your opinion, how risky is drowsy driving? Would you say that it is extremely risky, very risky, somewhat risky, not too risky, or not at all risky?") and the economy (e.g., "We are interested in how people are getting along financially these days. Would you say that you [and your family living there] are better off or worse off financially than you were a year ago?"). The experimental (p. 432) questions were administered twice, initially with standardized techniques and subsequently with either standardized or conversational techniques that allowed clarification about the labels on the response scale (e.g., "By 'extremely risky' we mean it causes an accident every time you do it. By 'not at all risky' we mean it never causes an accident") or about the key concept in the question (e.g., "By 'better off financially' we mean that you and your family have more left over at the end of each month than you did a year ago. Or, if you're spending more than you're earning, that you're adding less debt each month than you did a year ago"). The critical variable was response change between initial and subsequent administration, mirroring the logic used in Conrad and Schober (2000). When the questions were readministered with conversational interviewing, responses changed 18.1 percent of the time versus 12.7 percent under standardized readministration.

The evidence is thus that across both interviewer- administered and automated self-administered surveys, involving both behavioral and opinion questions, the ability to ground meaning through additional dialogue measurably improves respondents' comprehension and ultimately response accuracy. The inability to ground meaning in strictly standardized surveys has corresponding harmful effects on comprehension and thus response accuracy.

Evidence About Comprehension in Survey Dialogue

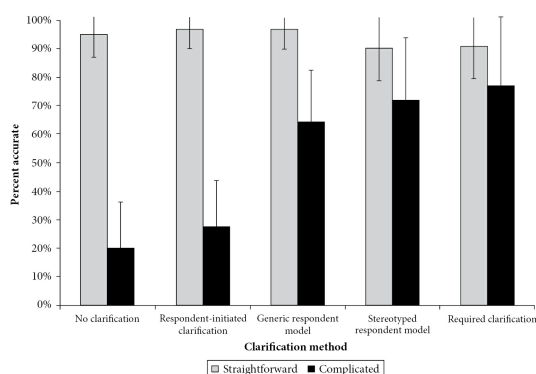
As these studies show, when interviewers actively diagnose respondents' need for clarification, they can improve response accuracy even more than when they rely only on respondents to explicitly request clarification, which in turn improves response accuracy relative to when they cannot provide clarification at all. This mirrors what happens in everyday conversations, in which people not only wait to be asked for clarification but also actively monitor each other closely for evidence of whether they are being understood and adjust what they say accordingly. The evidence people rely on is not only verbal but paralinguistic and visual, and it is used

systematically enough that, as we will see, interviewers and automated interviewing systems can potentially make use of it. But people also vary in how sensitive they are in everyday conversations to evidence of each other's comprehension, and they vary in how they indicate their own state of comprehension: whether they are prone to explicitly asking for clarification as opposed to relying on others to detect their need for clarification and when they produce which kinds of cues. This variability is also present in survey interviews.

In survey interviews and interactions with automated interviewing systems, many respondents do not explicitly request clarification even when other evidence suggests that they need it (Conrad & Schober, 2000; Schober & Conrad, 1997). They probably are relying on what Clark and Schober (1992) called the "presumption of interpretability": they assume that researchers have designed questions with them in mind and that if the researchers meant something different by a question than what it appears to the respondent to mean, then the researcher would have qualified the question. Nonetheless, respondents have been shown to display paralinguistic and visual evidence that they need clarification even when they don't explicitly request it.

For example, in an analysis of the laboratory telephone interviews from Schober and Conrad (1997), in which respondents answered questions about fictional scenarios so that the researchers could know when respondents were encountering a complicated mapping situation that was likely to require clarification, respondents' answers were more likely to include a filler (an *um* or *uh*) in answers to questions about complicated scenarios than in answers about straightforward scenarios (Schober & Bloom, 2004). Schober, Conrad, Dijkstra, and Ongena (2012) replicated this finding and also extended it to examine visual evidence of need for clarification in a corpus of Dutch face-to-face and telephone interviews in which students answered questions about their own lives rather than fictional scenarios. Answers that were disfluent (that included fillers and other disfluencies) and answers during which respondents looked away from the interviewer (what has been called "gaze aversion") were significantly more likely to be unreliable—that is, to change within a question-answer sequence in the interview or to change when respondents were provided with definitions for survey terms post interview. In both studies, the interviewers' ability to provide clarification (i.e., in administering conversational vs. standardized interviews) affected respondents' likelihood of providing disfluent answers and answers with gaze aversion. That is, while answering in conversational interviews, respondents were more likely to be disfluent and avert their gaze from the interviewer when their answers turned out to be unreliable. This demonstrates that the cues of need (p. 433) for clarification that a respondent produces can be affected by the interviewer's ability to respond to those cues, highlighting the depth of interactivity at play in survey dialogue.

Similar kinds of interactivity are also at play in dialogue with automated survey systems. In the Conrad, Schober, and Coiner (2007) study, in which laboratory respondents interacted with a browser-based (i.e., self-administered) questionnaire, respondents' delays in answering were informative about the extent to which they needed clarification. Conrad et al. hypothesized that respondent inactivity (no typing, no clicking) in this situation is likely to reflect comprehension difficulty, and they tested this by programming the survey interface to provide unsolicited clarification when respondents' inactivity exceeded a question-specific threshold (based on median response times for complicated scenarios in control conditions). In experiment 2, the thresholds were set differently for younger and older respondents, based on evidence from the cognitive aging literature strongly linking aging with slowing of behavior and mental processing. The evidence in both experiments was that response accuracy improved when respondents whose inactivity exceeded the thresholds were provided with unsolicited clarification; group-based modeling with different thresholds for different respondents improved accuracy more than a single threshold for all respondents.



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Fig. 26.2 Response accuracy for straightforward and complicated scenarios across five interviewing conditions: (1) no clarification (standardized), (2) respondent-initiated clarification, (3) system-initiated clarification delivered after generic pause duration (respondents can also request clarification), (4) system-initiated clarification delivered after stereotyped pause duration, i.e., different durations for younger and older groups (respondents can also request clarification), and (5) required clarification, delivered automatically after every question for all respondents.

These findings were mirrored in the Ehlen et al. (2007) speech dialogue system study, in which, in addition to pauses (the spoken equivalent of web browser inactivity), the researchers modeled other potential speech indicators of need for clarification, including fillers, restarts, hedges, and the like. Respondent pauses emerged as the best predictor of need for clarification—but with a twist: not only response times that were particularly slow but also response times that were particularly fast predicted inaccurate answers. This led them to identify an ideal range of response times for each question, within which answers were reliably more likely to be accurate. They demonstrated (see Figure 26.2) that when the speech dialogue interviewing system embodied these “just right” ranges (with different ranges for younger and older respondents), then providing unsolicited clarification when answers were too fast or too slow reliably improved response accuracy—in fact, just as much as providing (p. 434) clarification all the time (which has the downside of being irritating and increasing interview duration unnecessarily for all respondents).

Together, these findings strongly support the view that respondents’ paralinguistic displays provide systematic evidence of comprehension difficulty, in both speech and text dialogue systems, and in much the same way that they do in human interviews. Furthermore, the very same display can mean different things about different respondents: the same delay in responding by one respondent (a younger one in both these studies) can be informative about the need for clarification in a way that the same delay by another respondent (an older one in these studies) is not. The same is likely to be true of any other paralinguistic or visual displays: a filler produced during an answer by a chronically disfluent respondent should be less informative than the same filler produced by a highly fluent respondent, and gaze aversion during an answer by a respondent who never looks at an interviewer should mean less than gaze aversion by a respondent who makes continual eye contact (see Conrad, Schober, & Dijkstra, 2007). Consistent with a view that takes the base rates of respondents’ displays into account, Coiner, Schober, and Conrad (2011) demonstrated in a browser-based lab study that an individual respondent’s general speed of processing (measured through the Wechsler Adult Intelligence scale [WAIS] subtests at the individual level, rather than based on broad age groups) can strongly predict respondents’ likelihood of requesting clarification and eventual response accuracy.

These studies demonstrate that an automated interviewing system’s ability to detect and make use of a respondent’s paralinguistic displays can affect eventual response accuracy. Are the same dynamics at work in human interviews, where interviewers rely on their judgment (rather than preprogrammed algorithms) about whether and when to intervene with clarification? The evidence (Hubbard et al., 2012) is that interviewers can vary in their nonverbal sensitivity, as measured by the Profile of Non-Verbal Sensitivity, or PONS test (Hall, Murphy, & Schmid Mast, 2006; Rosenthal, Hall, Archer, DiMatteo, & Rogers, 1979) and that this affects their interviewing behavior and ultimately respondents’ accuracy. Interviewers who scored higher on the PONS were more efficient at communicating the intended meaning of questions in conversational interviews, taking less time and using fewer words. This is probably because they were less likely to engage in “preemptive strikes” (i.e., providing definitions before a respondent had a chance to indicate confusion or misinterpretation) than were conversational interviewers who scored lower on this attribute. Presumably, interviewers who are more sensitive to a speaker’s (in this case a respondent’s) nonverbal cues administer help more judiciously and efficiently, whereas less sensitive interviewers do not discriminate as well between responses that do and do not indicate the need for clarification.

Conclusion

Our argument is that pragmatic processes are at play in survey interviews to a far greater extent than most survey practitioners or researchers (let alone journalists or members of the general public) recognize. These processes have substantial effects on how respondents interpret questions, how survey dialogue unfolds, and what researchers should make of survey findings. In particular, we have reviewed evidence that:

1. Survey respondents—as they do in everyday conversation—make use of contextual and pragmatic information to infer a question's intended meaning.

When respondents answer, they rely on apparently formal features of the questionnaire, the content of preceding questions, and what they have answered already to interpret the questions. Unlike researchers, they may not treat the questions as independent, and they may take into account what they judge to be the researchers' intentions (e.g., based on the researcher's affiliation). In other words, they bring to the survey interaction their full interpretive capacities from everyday conversation, even when the inferences this leads them to make are (from the researcher's perspective) unwarranted. Also, as in everyday conversation, survey respondents are likely to assume that their interpretation of a survey question is the same as the researcher's, even though this is not necessarily the case (in everyday conversation or surveys). The question that a respondent answers may not be the question that the researcher wanted to ask, nor do the answers provided by different respondents necessarily pertain to the same behavior.

2. Survey interviews involve more than simply question asking and answering.

Pragmatic processes are involved not only in respondents' immediate interpretation of the question, but in subsequent dialogue, whether that dialogue is strictly standardized or more collaborative. (p. 435) How respondents answer questions—how long they take to answer, how fluent their answers are, whether they request clarification or not—is informative about the accuracy and reliability of their answers. This all stems from the fact that the back-and-forth of survey dialogues is related to the back-and-forth of everyday conversation, in which people assume that they can work with each other collaboratively to ground their understanding. They use these skills to interpret what happens even in strictly standardized interviews that are not intended to allow grounding and clarification; survey respondents try to make sense of the questions they are asked using whatever conversational resources they can bring to bear.

It is certainly not our intention to suggest that all survey research is flawed because of the kinds of pragmatic processes in survey interviews that we are reviewing. But we propose that understanding what happens in survey responding requires more careful attention to how the questions are presented and how the answers are produced in dialogue. The implication for researchers is that they would be wise to assume that respondents are engaged in all the pragmatic processes we have reviewed and to design questionnaires and interviewing procedures that build on them, rather than ignoring them or wishing they were not present.

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Notes:

(1) . For example, web surveys can allow respondents to request clarification with a mouse (e.g., Conrad, Schober, & Coiner, 2007), and speech dialogue interviewing systems (speech-IVR, interactive voice response) can, in principle, recognize spoken requests for clarification or even diagnose paralinguistic evidence of confusion or misunderstanding (see Schober, Conrad, & Bloom, 2000; Ehlen, Schober, & Conrad, 2007).

Frederick G. Conrad

Frederick G. Conrad is Research Professor and Director, Michigan Program in Survey Methodology, University of Michigan and Joint Program in Survey Methodology, University of Maryland.

Michael F. Schober

Michael F. Schober is Professor of Psychology at the New School for Social Research.

Norbert Schwarz

Norbert Schwarz is Provost Professor in the Department of Psychology and Marshall School of Business at the U of Southern California.

